VTX User Manual

Exploring and editing a procedural universe with the Virtual Terrain Explorer

Dean Sindorf

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Part I: User guide

Introduction

VTX (the Virtual Terrain Explorer) is a program for exploring a procedurally generated universe, from whole galaxies down to the pebbles on a planet's surface. Nothing in a VTX universe is stored as a model or a map. Planets, clouds, rocks and plants are computed from short expressions while you watch, and each one gets more detailed as you get closer.

You can use VTX in three ways:

- **Explore.** Fly between stars, orbit planets and walk across their surfaces.
- **Edit.** Change any object through the Properties dialog, from the height of a mountain range to the color of a sunset, and see the result immediately.
- **Create.** Generate new stars, planets and moons with a single command, then refine them and save them as scenes or reusable objects.

This manual explains how to operate VTX. Part I is a guide to installing the program and to the tasks you will do most often. Part II is a reference to every menu, key and dialog control. Part II is generated from the program's source code, so it always matches the version of VTX it ships with.

How VTX works internally is covered in the separate VTX Theory of Operation document.

Conventions

- Menu commands are written as **File > Open**.
- Keys are written as they are typed. VTX's keyboard commands are case-sensitive, so **s** and **S** are different commands.
- Values with units in the dialogs use the units shown in the control's label, for example *Day(hrs)* or *Ht(mls)* (miles).

Installing and starting VTX

System requirements

- Windows 10 or 11, 64-bit.
- A graphics card and driver with OpenGL support for GLSL and geometry shaders. A dedicated graphics card is strongly recommended.
- About 1 GB of free disk space, more if you generate many worlds or record movies.

Installing

VTX is distributed in two forms:

- **Installer** (*VTX-win64-version-setup.exe*). Run it and follow the prompts. VTX is installed for the current user in *%LOCALAPPDATA%\Programs\VTX*, and no administrator rights are needed. The installer adds VTX to the Start menu, optionally to the desktop, and to Settings > Apps for uninstalling.
- **Zip archive** (*VTX-win64-version.zip*). Unpack it into any folder you can write to, for example your Documents or Downloads folder.

> **Important:** VTX writes into its own folders while it runs: generated shader files, saved scenes, preferences and generated textures. Don't install or unpack it under *C:\Program Files*, where ordinary users can't write.

Starting VTX

Start VTX from the Start menu entry or desktop icon (installer), or by double-clicking **VTX.bat** in the VTX folder (zip). Don't start *bin\vtx.exe* directly from Explorer: VTX must start with the *bin* folder as its working directory, and the shortcut and VTX.bat take care of that.

When VTX quits it saves your session as *Saves\last.spx*. The next time you start it the same way, it reopens that session. The first time, it opens the default scene instead.

To pin VTX to the taskbar when using the zip version, create a shortcut whose target is `C:\Windows\System32\cmd.exe /c "C:\path\to\VTX\VTX.bat"`, set its **Run** option to *Minimized* and its icon to *bin\vtx.exe*, then right-click it and choose **Pin to taskbar**.

Command-line options

VTX.bat passes its arguments to vtx.exe:

Argument	Effect
`--s`, `--m`, `--b`	Small, medium (default) or big window
`--w`, `--p`	Wide or panoramic window

Argument	Effect
name	Open scene *name* from the Saves folder, for example `alpine`
`-`*key*	Apply a single-key command at startup, for example `-s` toggles shadows

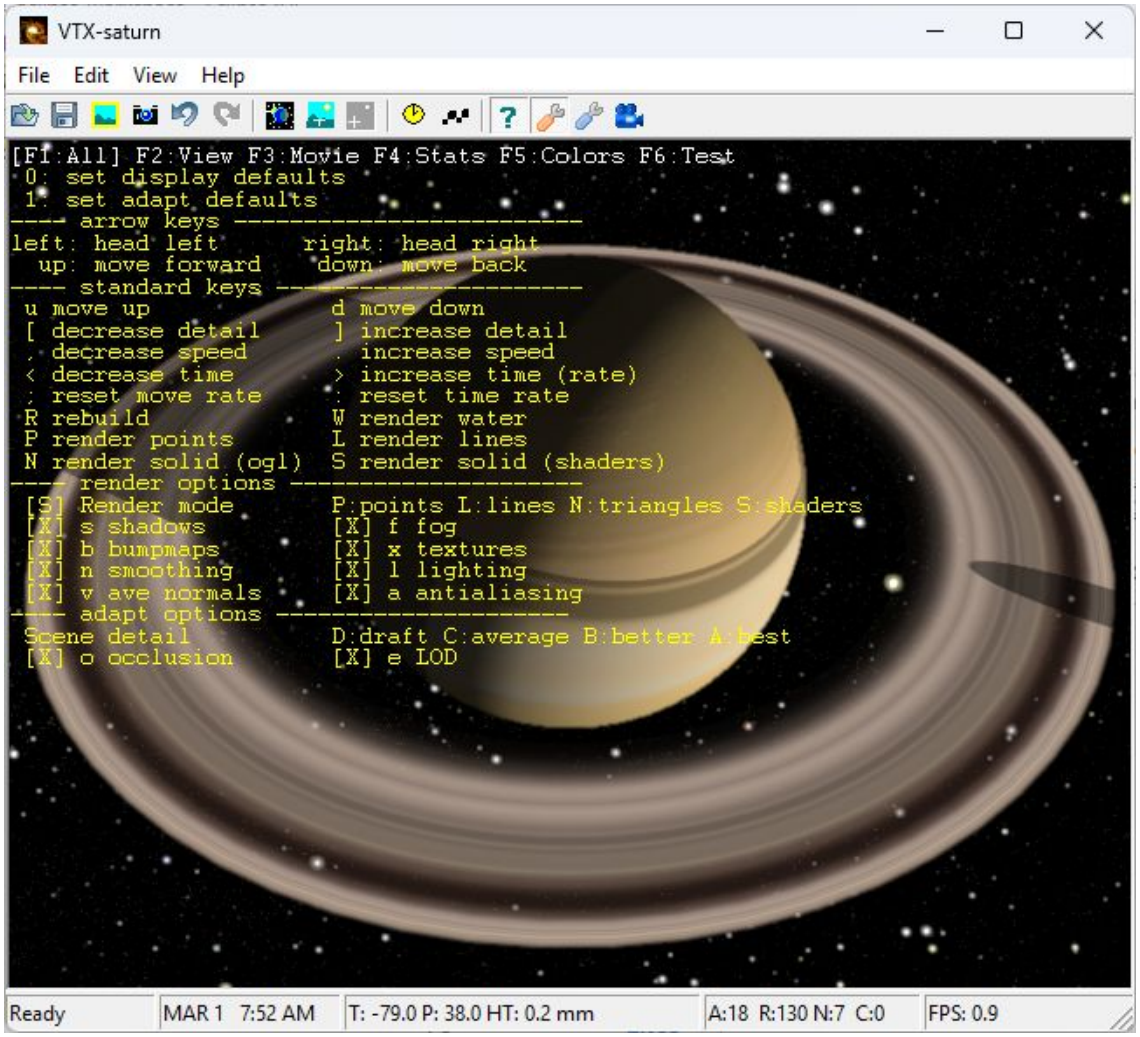
The VTX folder

Folder	Contents
bin	The program
Docs	This manual and the Theory of Operation
Saves	Scene files (*.spx*), including *last.spx* and *Default.spx*
Objects	Saved objects, by type (a new installation includes only plants)
Textures	Image textures, plant images and sprite images
Bitmaps	Textures that VTX generates for new worlds (see Generating new worlds)
Images	Snapshots saved with **File > Save Image**
Movies	Movie frames
Resources	Star and cloud sprites, and saved preferences
Shaders	The program's GPU shaders

Graphics card selection

Many laptops have two graphics processors: a power-saving one built into the processor and a faster dedicated one. VTX asks Windows for the dedicated graphics card. If pictures look coarse or slow, check which one is in use: VTX prints it in its console window when it starts, on a line beginning *OpenGL:*. You can also set it in **Settings > System > Display > Graphics**: add *bin\vtx.exe* and choose **High performance**.

The main window

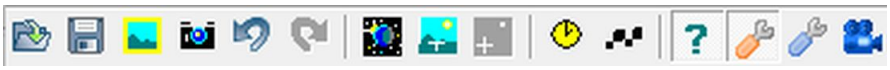


The main window, with the on-screen key help (F1) showing

The main window shows the scene, with a menu bar and toolbar above it and a status bar below it.

The toolbar

The toolbar gives one-click access to the commands you use most. Rest the pointer on a button to see its name and what it does. This manual refers to the buttons by the names below.



The toolbar, with Key Help on. Redo and Selection are grayed out because there is nothing to redo or select.

From left to right, the buttons fall into four groups: files and view history, moving, animation, and help and dialogs.

	Name	What it does	Same as
	**Open**	Opens a scene file (.spx) and builds it. Scenes are kept in the *Saves* folder.	File > Open
	**Save**	Saves the current scene, including changes made in the Properties dialog, to a scene file (.spx).	File > Save
	**View Image**	Shows a saved JPEG or BMP image, such as a snapshot, in the main window.	File > View Image
	**Snapshot**	Saves the current view as a JPEG or BMP image. You choose the file name; the *Images* folder is the default.	File > Save Image
	**Undo**	Goes back to the previous view in the view history. Grayed out when there is no earlier view.	Edit > Undo, **-**
	**Redo**	Goes forward to the next view in the history, after an Undo. Grayed out when there is no later view.	Edit > Redo, **+**
	**Orbit**	Moves out to an orbital view of the current object, so you see the whole world. With a star selected, takes you to that star's system.	View > Move to > Orbit, **Shift+U**
	**Surface**	Moves down to the surface of the current object. If you have marked a point with the white crosshair, takes you there.	View > Move to > Surface, **Shift+G**
	**Selection**	Moves to the point marked with the white crosshair, keeping your current height. Grayed out when nothing is selected.	View > Move to > Selection, **Shift+M**
	**Time**	Starts or stops time: planets rotate and orbit and clouds drift. The button stays pressed while time runs.	View > Animate > Time
	**Motion**	Starts or stops motion: the view keeps moving at the current speed. The button stays pressed while it is on.	View > Animate > Motion
	**Key Help**	Shows or hides the list of keyboard commands over the scene. The button is pressed while the list shows.	Help > key help, **?**
	**Properties**	Opens the Properties dialog, where you inspect and edit the objects in the scene.	**Tab**
	**Image Editor**	Opens the Image Editor, for creating and editing the images used as textures.	
	**Movie Editor**	Opens the Movie Editor, for recording and playing paths and movies.	

Buttons that turn something on, such as **Time**, **Key Help** and the three dialog buttons, stay pressed while that thing is on or open. Click again to turn it off or close it. **View > Show Toolbar** hides and shows the toolbar.

On-screen information and key help

VTX draws status information over the scene. Press **?** (or click the **Key Help** toolbar button) to show or hide a list of the keyboard commands available in the current mode.

The function keys **F1** to **F6** switch between modes. Each mode gives the number keys **0** to **9** a different meaning:

Key	Mode	Number keys
F1	All (default)	Display and detail defaults
F2	View	Turn, look and tilt the view
F3	Movie	Record and play paths and movies
F4	Stats	Show statistics about objects, maps and rendering
F5	Colors	Color the terrain by internal properties (diagnostic)
F6	Test	Alternative views for testing

The full list for each mode is in Part II, Keyboard.

Moving around

View types

VTX has three ways of moving, which you choose with **View > Move to**, the toolbar or the keyboard:

- **Orbit** (**Shift+U**). The view circles the current object, typically a planet, at a distance. Use it to look at a whole world. If a star is selected, Orbit takes you to that star's system instead (see Traveling to the stars).
- **Surface** (**Shift+G**). The view moves down to the surface of the current object, as if you were walking or flying above the ground. If you have marked a point with the crosshair, Surface takes you there.
- **Selection** (**Shift+M**). The view moves to the point marked with the crosshair and keeps its current height.

Keyboard movement

Key	Action
Up / Down arrow	Move forward / back
Left / Right arrow	Turn left / right

Key	Action
u / d	Move up / down
Ctrl+u / Ctrl+d (also Ctrl+Up / Ctrl+Down)	Tilt the view up / down
, / .	Decrease / increase the speed of movement
;	Reset the speed of movement

In surface view, the F2 (View) mode adds keys to look and turn in fixed steps, and **9** and **0** switch between an overhead view and a normal surface view.

Mouse movement

Mouse	Orbit view	Surface view
Left-drag sideways	Circle around the object	Turn
Left-drag up or down	Move closer or farther	Move forward or back
Right-drag sideways	Circle around the object	Look to the side
Right-drag up or down	Tilt the view	Look up or down
Wheel	Zoom in or out	Lower or raise the viewpoint
Hold middle button	Fly forward while held	Fly forward while held

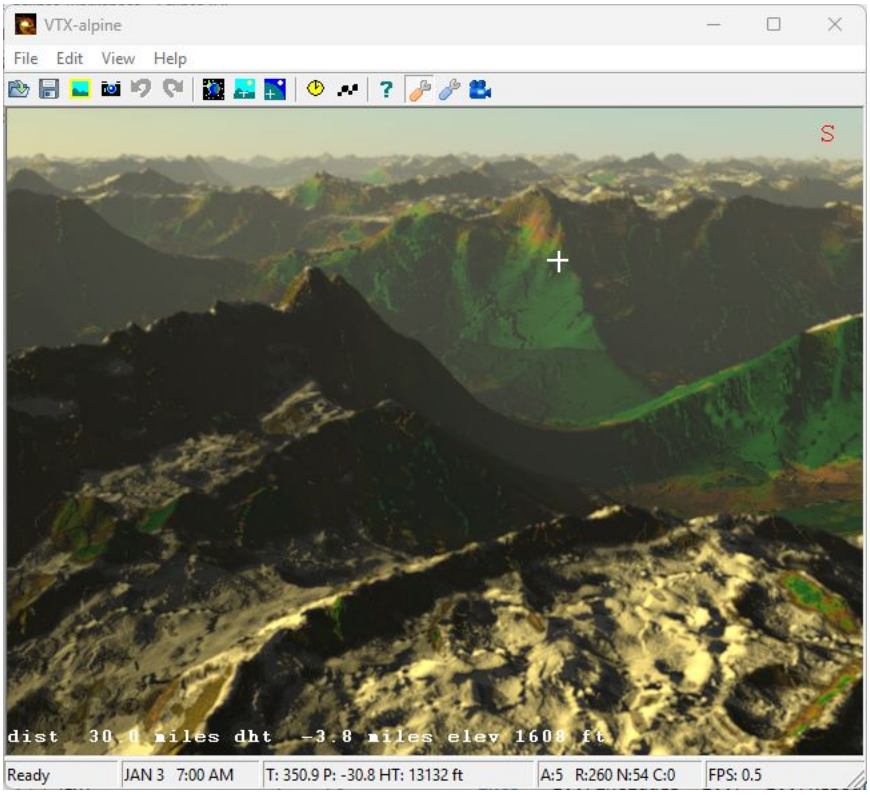
Choosing where to go

Click the surface of a planet or moon, in orbit or surface view, and a white crosshair marks the point you clicked. A line of text at the bottom of the view shows the distance to the point (*dist*) and its elevation (*elev*).

Once a point is marked:

- **Surface** takes you to the crosshair and lowers you to the ground.
- **Selection** takes you to the crosshair but keeps your current height, so you see the spot from the same altitude.

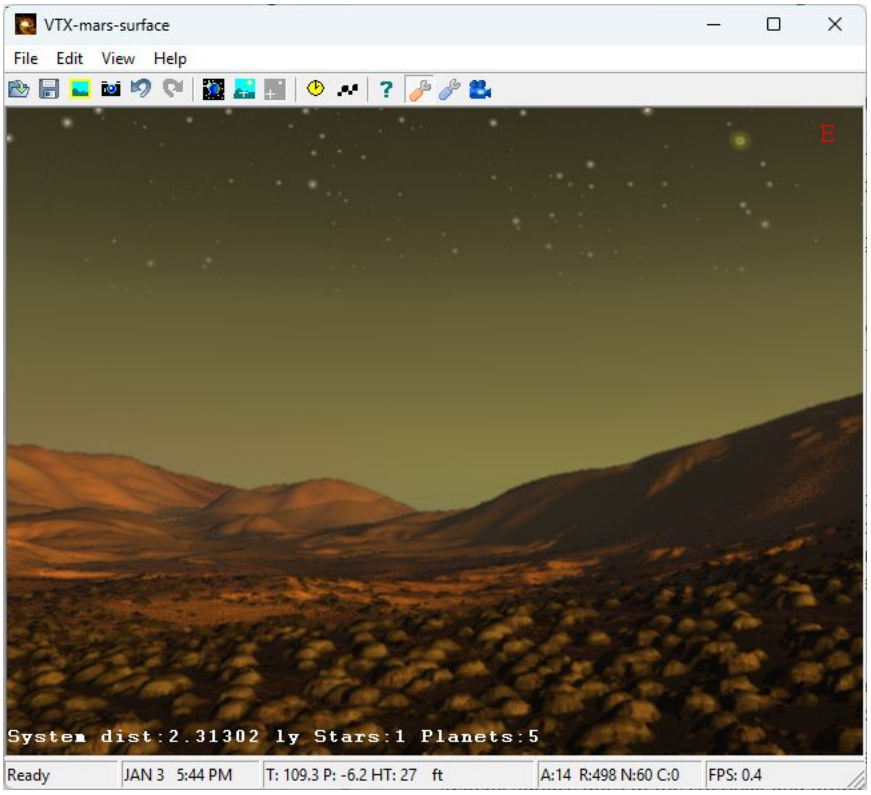
Both work the same way from orbit view and from surface view. Clicking a planet or moon also selects it in the Properties dialog. Press **Space** to clear the selection.



The crosshair marks a point on the surface; its distance and elevation show at the bottom

Traveling to the stars

Click a star in the star field and it turns yellow to show that it is selected. Only the nearer, foreground stars can be selected; the background stars can't. A line of text at the bottom of the view shows the distance to the star's system and how many stars and planets it has, for example *System dist:2.31302 ly Stars:1 Planets:5*.



A selected star (yellow, upper right) and its system's distance, stars and planets at the bottom

Then click **Orbit** (or press **Shift+U**). VTX transfers you to that system, with the system selected. When you arrive at a star that has never been visited, VTX generates its planetary system on the spot, so you can explore the universe one star at a time.

View history

VTX remembers the views you visit, so you can return to them:

Key	Action
=	Mark the current view
- / +	Go to the previous / next view in the history
_	Rewind to the first view
z	Undo the last move
Z	Clear the view history

Edit > Undo and **Edit > Redo** (and the toolbar buttons) step through the same history. **View > Clear History** clears it.

Detail, quality and speed

VTX keeps refining what you see until the scene is as detailed as the current settings allow. A scene often

sharpens over a second or two after you stop moving.

Scene quality

View > Scene Quality sets the overall level of detail:

Setting	Key
Low	SHIFT-D
Medium	SHIFT-C
High	SHIFT-B
Very High	SHIFT-A

] and **[** (**More Detail** and **Less Detail**) adjust the detail in smaller steps. Higher detail looks better but renders more slowly, so lower it while moving fast and raise it for pictures.

Render options

The keyboard toggles rendering features on and off. The most useful are:

Key	Feature
s	Shadows
f	Fog
x	Textures
b	Bump maps
l	Lighting
W	Water
a	Anti-aliasing
m	High dynamic range
S / N / L / P	Draw shaded (normal), solid, lines or points
R	Rebuild the scene

With key help showing (**?**), the F1 list shows which features are on (**[X]**) and off (**[ ]**). The Scene object's tabs in the Properties dialog give finer control over the same settings.

Rebuild (**R**, or **Edit > Rebuild**) regenerates everything in view. Use it if the display looks wrong after many edits.

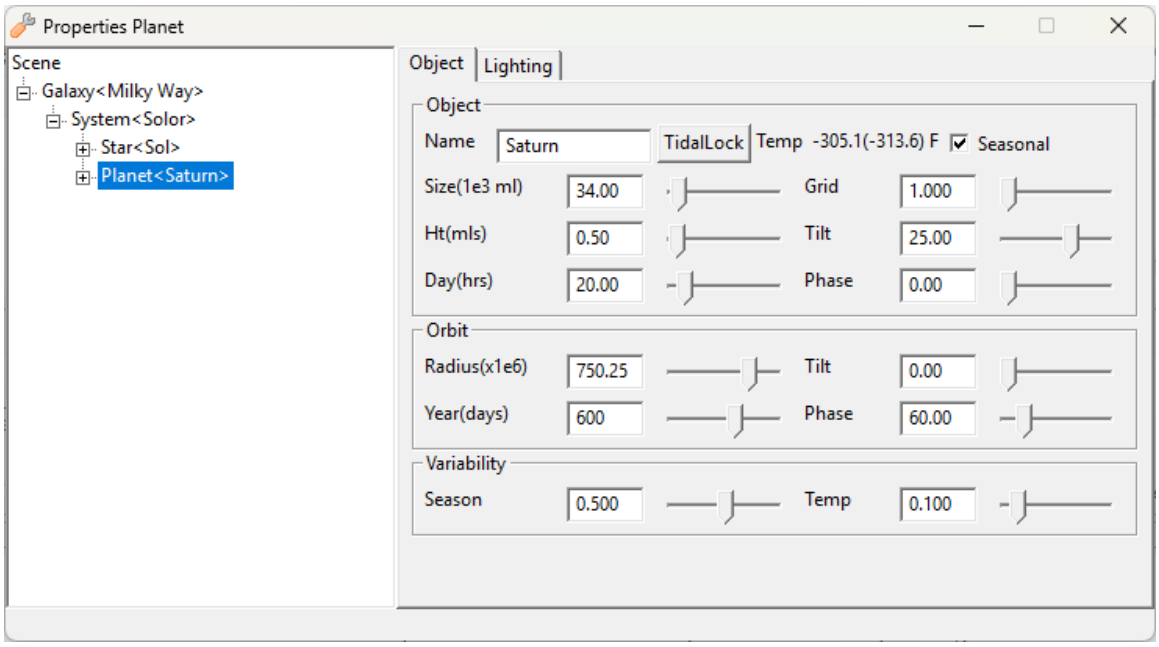
Time and animation

Planets and moons rotate and orbit, and clouds drift. Time normally stands still, so you can study a view. To make time run:

- The **Time** toolbar button (or **View** > **Animate** > **Time**) starts and stops the clock.
- < and > decrease and increase the time rate. **:** resets it.
- The **Motion** toolbar button (or **View** > **Animate** > **Motion**) makes the view keep moving at the current speed, so you can glide over a surface hands-free.

The Scene object's **Display** tab shows the time and has a multiplier from x1 to x10^7 for fast-forwarding orbits.

The Properties dialog



The Properties dialog: the object tree on the left, a Planet's Object tab on the right

The Properties dialog is where you inspect and change a scene. Open it with the **Properties** toolbar button or the **Tab** key.

The object tree

The left side shows the scene as a tree, which mirrors how the universe nests: the Scene contains a Galaxy, which contains a System, which contains a Star and Planets, and a Planet contains its Moons, Sky, Clouds, Rings and a Surface with its layers and features.

Click an object in the tree to show its tabs on the right. Selecting an object in the view also selects it in the tree.

Changing values

Each object type has its own set of tabs. Part II lists every tab and control. The controls work the same way everywhere:

- **Sliders** have a text box beside them. Drag the slider, or type a value in the box and press **Enter**.
- **Expression sliders** accept either a number or an expression, so a value can vary across a surface instead of being constant.
- **Color sliders** set a color with a color box and its intensity with a slider.
- **Expression fields** hold a complete terrain or color expression. Edit the text and press **Enter** to apply it.

Most changes appear in the view immediately. Changes that affect terrain shape take a moment while the surface is rebuilt.

Editing parts of an expression

Expressions in a surface can be long. VTX can show the noise, fractal, crater and erosion functions in an expression as short tokens such as *Noise1*, and edit each one in its own dialog:

1. Double-click a token in an expression field to select it.
2. Right-click and choose **Edit**. The Noise or Function editor opens with that function's settings.
3. Change the settings. The expression updates when you apply them.

To insert a new function, right-click in the field with nothing selected and choose **Add > Noise**, **Fractal**, **Craters** or **Erode**.

The object menu

Right-click an object in the tree for its menu. What appears depends on the object type. The most common commands are:

Command	Action
Set View	Make this object the center of the view
Show	Show or hide the object
Enable	Turn a feature (such as fog or erosion) on or off without deleting it
Delete	Remove the object from the scene
Add	Add a child object: a moon to a planet, a cloud layer, a plant, and so on
Replace with..	Replace the object with a saved one or a newly generated one
Save..	Save the object to the Objects library so you can reuse it
Randomize	Give the object new random settings of the same kind
Default	Reset the object to its default settings
Remove file..	Delete a saved object from the Objects library

You can also rearrange the tree by dragging objects with the mouse. Dragging with the right mouse button

copies an object instead of moving it.

Generating new worlds

Every star you travel to gets a generated planetary system. You can also generate objects deliberately:

- **Add** > ***type*** > **Generate** adds a new generated object, for example a new moon around a planet.
- **Replace with..** > **Generate** replaces an object with a newly generated one of the same type. Some types offer sub-types, for example rocky, icy or gas giant planets.
- **Add** or **Replace with..** followed by **<Random>** picks one of your saved objects of that type at random.
- **Randomize** keeps the object's type but gives it new random settings.

If you don't like a result, generate again. Each generation is different.

Reusing generated textures

Generated worlds use texture images: height maps, erosion maps, layer images and color bands. Creating them takes time, so VTX can save them in the *Bitmaps* folder and reuse them for later worlds. Two settings on the Scene object's **Adapt** tab, in the *Procedural Generation* group, control this:

Setting	Choices
Reuse	**All**: reuse saved images for everything. **2D** (default): reuse saved 2D maps, create new color bands. **None**: create every image new.
Keep	Which generated images to keep in *Bitmaps*: **All**, **2D** (default) or **None**. Images that aren't kept are deleted the next time VTX starts.

With a new installation, *Bitmaps* is empty, so the first worlds you generate create all their own images. As *Bitmaps* fills up, generation gets faster and the 2D maps are drawn from a growing collection.

Scenes and saved objects

Opening and saving scenes

A scene is everything in the universe you are exploring, plus the views you have saved. Scenes are stored as text files with the extension *.spx* in the *Saves* folder.

- **File** > **Open** opens a scene.
- **File** > **Save** saves the current scene.
- VTX saves the current session to *last.spx* when it quits, and reopens it the next time it starts.

Scene files are plain text that describes the object tree and its expressions, so they can also be read and edited

in a text editor. The Theory of Operation describes the format.

Saved objects

Any object can be saved on its own with **Save..** in its tree menu. Saved objects go into the *Objects* folder, organized by type, and then appear in the **Add** and **Replace with..** menus of matching objects. This is the way to build a library of favorite planets, cloud layers, plants or rock types and reuse them across scenes.

A new installation includes only the plants in *Objects\Plant*, which **Generate** draws on when it adds grass, bushes and trees to a world. The other folders are empty, so until you save objects of your own, the **Add** menus offer only the built-in default for those types. Scenes don't depend on the library, because a scene file contains all of its objects.

The VTX repository on GitHub (https://github.com/dwsindorf/VTX) has a collection of example objects in its *Objects* folder. To use them, copy the .spx files you want into the matching subfolder of your installation, for example *Objects\Planet*. Some were saved with earlier versions of VTX, so they may not load, or may look different from when they were made. Some also use textures that VTX generates into the *Bitmaps* folder, which a new installation doesn't have yet.

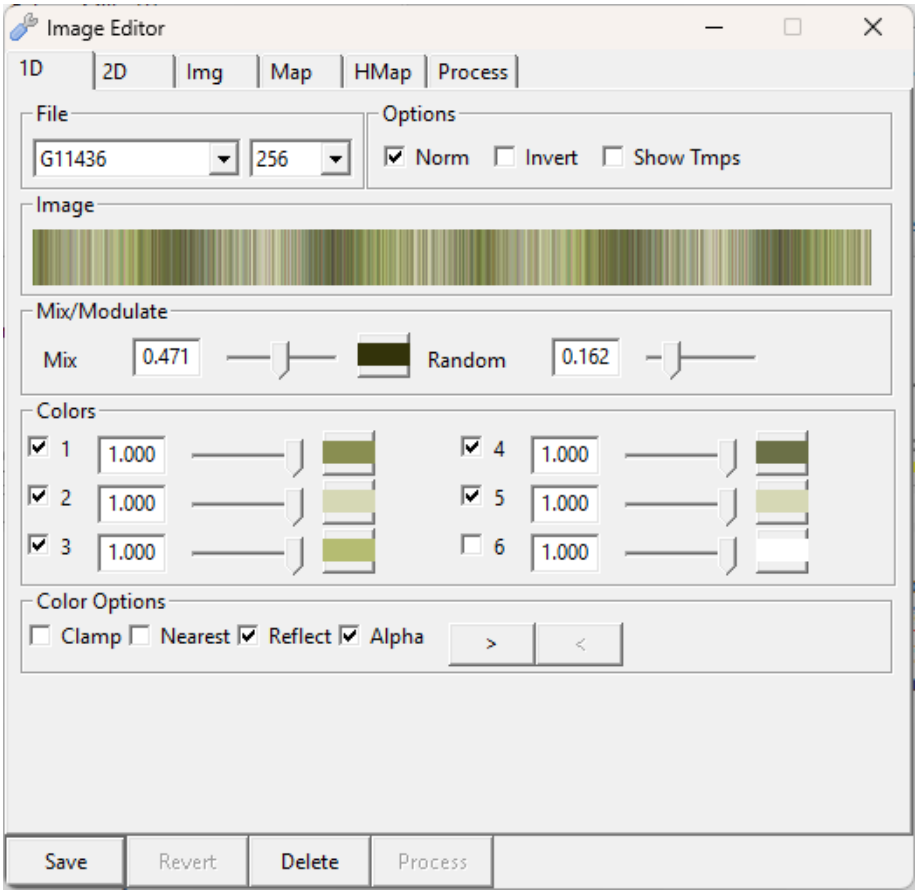
Pictures and movies

Snapshots

File > Save Image (the **Snapshot** toolbar button) saves the current view as an image in the *Images* folder. **File > View Image** opens a saved image.

For the best result, raise the scene quality and let the view settle before you save.

The Image Editor

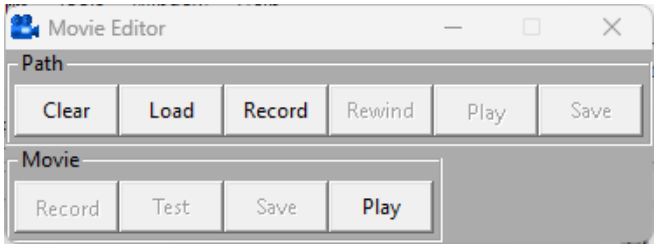


The Image Editor, showing the 1D (bands) tab

The Image Editor creates and edits the images that surfaces use as textures. Open it with the **Image Editor** toolbar button. Its tabs are:

Tab	Purpose
1D	Color bands: one-dimensional color gradients, used for layered and striped surfaces
2D	Two-dimensional images generated from expressions
Img	Imported images from *Textures\Images*
Map	Imported surface maps
HMap	Imported height maps
Process	Image processing (for example blurring) of images in *Textures\Processed*

Movies



The Movie Editor

VTX records a movie in two steps. First you record a *path* by flying through the scene. Then VTX replays the path and saves a frame at each step. Recording frames from a path gives smooth movement even when rendering each frame is slow.

Press **F3** to switch the number keys to Movie mode:

Key	Action
2	Start recording a path. Move around the scene as you like.
1	Stop
3 / 4	Rewind / play the path
5 / 6	Save / load the path
7	Record movie frames from the path
8	Record movie frames live, as you move
0	Clear the path and movie

The **Movie Editor** toolbar button opens the Movie Editor, which has the same commands as buttons: **Clear**, **Load**, **Record**, **Rewind**, **Play** and **Save** for paths, and **Record**, **Test**, **Save** and **Play** for movies. Frames are recorded as images in the *Movies* folder.

Save turns the recorded frames into a video file, and **Play** plays one back. Both use the free FFmpeg tools (*ffmpeg* and *ffplay*), which aren't included with VTX: install FFmpeg and make sure its *bin* folder is on your PATH. In the Play dialog, **Exit at end** closes the player when the video finishes.

Troubleshooting

The picture is coarse or slow. Check that VTX is using the dedicated graphics card (see Graphics card selection). Lower the scene quality while moving.

Parts of the Properties dialog are cut off. VTX's dialogs are laid out for 100% display scaling. If Windows display scaling (**Settings > System > Display > Scale**) is 125% or more, some controls may not fit. Setting the scale to 100% avoids this.

VTX shows a shader error in its console. The console lists the shader that failed. Make sure the *Shaders* folder is complete and writable. VTX writes a *.debug* copy of each shader there, which shows exactly what

was sent to the graphics driver.

Settings seem wrong after experimenting. Press **0** in F1 mode to restore the display defaults, and **1** to restore the detail (adapt) defaults. Saved preferences are kept in *Resources\UI\prefs.spx*.

The scene looks wrong after many edits. Press **R** to rebuild it.

Part II: Reference

This part is generated from the VTX source code (v1.0-1-g1562c8f-dirty), so it matches this version of the program exactly. Ranges are the ranges of the sliders: you can often type values outside them.

Menus and toolbar

File menu

Command	Key	Notes
Open ..		
Save ..		
Save Image ..		
View Image ..		
Exit		

Edit menu

Command	Key	Notes
Undo Z		
Redo Y		
Rebuild	SHIFT-R	

View menu

Command	Key	Notes
Show Toolbar		Show/Hide the toolbar
Clear History		
Move to		
Move to > Orbit	SHIFT-U	
Move to > Surface	SHIFT-G	

Command	Key	Notes
Move to > Selection	SHIFT-M	
Animate		
Animate > Time		
Animate > Motion		
Window Size		
Window Size > Small		
Window Size > Medium		
Window Size > Big		
Window Size > Wide		
Window Size > Panoramic		
Scene Quality		
Scene Quality > Low	SHIFT-D	
Scene Quality > Medium	SHIFT-C	
Scene Quality > High	SHIFT-B	
Scene Quality > Very High	SHIFT-A	
Scene Quality > More Detail	]	
Scene Quality > Less Detail	[	

Help menu

Command	Key	Notes
key help ?		Show/Hide Onscreen key help
About...		Show about dialog

Toolbar

Button	Action
Open	load a scene file (.spx)
Save	save the scene to a file (.spx)
View Image	show a saved JPEG or BMP image
Snapshot	save the current view as an image
Undo	go back to the previous view (-)
Redo	go forward to the next view (+)
Orbit	view the current object from orbit, or go to the selected star (Shift+U)
Surface	go down to the surface, at the crosshair if one is set (Shift+G)
Selection	go to the crosshair, keeping the current height (Shift+M)

Button	Action
Time	start or stop time animation
Motion	start or stop moving at the current speed
Key Help	show or hide the on-screen key list (?)
Properties	open the Properties dialog to edit the scene (Tab)
Image Editor	create and edit image textures
Movie Editor	record and play paths and movies

Keyboard

Press **?** to show the on-screen key help, and **F1** to **F6** to switch modes. The number keys **0** to **9** do different things in each mode. Letter keys are case-sensitive.

F1: All mode

Key	Action
0	Set display defaults
1	Set adapt defaults
Arrow keys	
left	Head left
right	Head right
up	Move forward
down	Move back
Standard keys	
u	Move up
d	Move down
[	Decrease detail
]	Increase detail
,	Decrease speed
.	Increase speed
<	Decrease time
>	Increase time (rate)
;	Reset move rate
:	Reset time rate
R	Rebuild
W	Render water
P	Render points

Key	Action
L	Render lines
N	Render solid (ogl) S render solid (shaders)
Render options	
Render	Mode
P	Points L:lines N:triangles S:shaders
s	Shadows
f	Fog
b	Bumpmaps
x	Textures
n	Smoothing
l	Lighting
v	Ave normals
a	Antialiasing
Adapt options	
Scene	Detail
D	Draft C:average B:better A:best
o	Occlusion
e	LOD

F2: View mode

Key	Action
1	Right turn
2	Left turn
3	Look left
4	Look right
5	Look down
6	Look up
7	Tilt left
8	Tilt right
9	Overhead view
0	Surface view (reset)

F3: Movie mode

Key	Action
0	Clear (all)
1	Stop
Path	
2	Record
3	Rewind
4	Play
5	Save
6	Load
Movie	
7	Record (from path)
8	Record (live)
9	Test

F4: Stats mode

Key	Action
0	Clear all
1	Objects
2	Maps
3	Cells
4	Raster
5	Stars
6	Placements
7	Terrain
8	Plants

F5: Colors mode

Key	Action
0	Normal colors
1	Node size
2	Node type
3	Node visibility
4	Node layers
5	Node edges

Key	Action
6	Node margins
7	Node tests

F6: Test mode

Key	Action
0	Normal view
1	Laboratory view
2	Light source view

All key commands

Every single-key command, as defined in the program. Commands marked *developer* are for testing and diagnostics.

Key	Command
F1	Key help mode: All (default)
F2	Key help mode: View
F3	Key help mode: Movie
F4	Key help mode: Stats
F5	Key help mode: Colors
F6	Key help mode: Test
F7	F7 (*developer*)
F8	F8 (*developer*)
F9	F9 (*developer*)
F10	F10 (*developer*)
Left arrow	Turn (head) left
Right arrow	Turn (head) right
Up arrow	Move forward
Down arrow	Move back
?	Show or hide the on-screen key help
!	Test1 (*developer*)
@	Test2 (*developer*)
#	Test3 (*developer*)
$	Test4 (*developer*)
%	Test5 (*developer*)
^	Test6 (*developer*)

Key	Command
&	Test7 (*developer*)
*	Test8 (*developer*)
Q	Make the default scene (at startup)
u	Move up
d	Move down
Ctrl+u	Tilt the view up (also Ctrl+Up)
Ctrl+d	Tilt the view down (also Ctrl+Down)
M	Move to the selected object
_	Rewind to the first saved view
-	Previous view in the history
+	Next view in the history
Space	Clear the selection
=	Mark (save) the current view
z	Undo the last move
Z	Clear the view history
I	New random seed
y	Yydebug (*developer*)
q	Quit
.	Increase movement speed
,	Decrease movement speed
>	Increase the time rate
<	Decrease the time rate
g	Start or stop continuous motion
t	Start or stop time
;	Reset movement speed
:	Reset the time rate
]	More detail
[	Less detail
G	Surface view
O	Orbit view
U	Orbit (global) view
V	Vertical view
H	Horizontal view
\	Raster shadow map (*developer*)

Key	Command
k	Raster lines (*developer*)
S	Draw shaded (shaders)
N	Draw solid (OpenGL)
L	Draw lines
P	Draw points
W	Water on/off
w	Water depth on/off
c	Mix (*developer*)
x	Textures on/off
X	Sprites on/off
Y	Specular light on/off
b	Bump maps on/off
r	Water reflections on/off
l	Lighting on/off
n	Smooth shading on/off
s	Shadows on/off
T	Twilight on/off
f	Fog on/off
v	Averaged normals on/off
i	Nvis (*developer*)
F	Face (*developer*)
a	Anti-aliasing on/off
m	High dynamic range on/off
j	Geometry shader terrain on/off
o	Occlusion test on/off
e	Level of detail on/off
E	Uniform (*developer*)
h	Hidden (*developer*)
R	Rebuild the scene
A	Scene quality: Very High
B	Scene quality: High
C	Scene quality: Medium
D	Scene quality: Low

Properties dialog

Each object type in the scene tree has its own tabs. For each type, this chapter lists its right-click menu, then every tab, grouped as in the dialog.

Scene

Menu: Add

Display tab

Control	Type	Range or choices	Notes
Animation			
Time	slider	1 to 10	
Animation	list	x1, x10, x100, x1000, x10^4, x10^5, x10^6, x10^7	
Animation	list	x1, x10, x100, x1000, x10^4, x10^5, x10^6, x10^7	
Rate	slider	1 to 10	
+	checkbox		
View			
FOV	slider	20 to 75	
Draw Mode	option buttons	Points, lines, Solid, Shaded	
Lighting Model	option buttons	OGL, Phong, Mix	

Adapt tab

Control	Type	Range or choices	Notes
Vertex Generation			
Detail	list	Draft, Normal, High, Photo	
CellSize	slider	1 to 10	
Geometry Level	list	1, 2, 3, 4, 5, 6	
Procedural Generation			
Detail	list	Low, Medium, High, Max	
Layers	list	Auto, 1, 2, 3, 4	
Reuse	list	All, 2D, None	Which saved generated images new worlds reuse (see Generating new worlds).
Keep	list	All, 2D, None	Which generated images are kept in the Bitmaps folder.
Adapt Tests			
Occlusion	checkbox		Decrease resolution in occluded areas

Control	Type	Range or choices	Notes
Clipping	checkbox		Decrease resolution in off-screen areas
Backfacing	checkbox		Decrease resolution in backfacing sides
Curvature	checkbox		Decrease resolution in smooth areas
Adapt Tests	option buttons	Overlap, Merge	

Render tab

Control	Type	Range or choices	Notes
Render Options			
HDR	checkbox		Enable High dynamic range
Shadows	checkbox		Enable shadows
Water	checkbox		Enable water
High Dynamic Range			
Min	slider	0.0 to 30.0	
Max	slider	0.0 to 10.0	
Shadows			
Blur	slider	0.1 to 2.0	
Views	list	1, 2, 3, 4, 5, 6, 7, 8	
Cascades	checkbox		Use cascades
FOV	slider	0.5 to 1.5	
DOV	slider	0.1 to 1.5	

Filter tab

Control	Type	Range or choices	Notes
Filter Options			
AveNorms	checkbox		Average normals at each vertex
Aniso	checkbox		Anisotropic texture filtering
Edges	checkbox		Apply image filter
Big Kernel	checkbox		use large filter kernel
Show Mask	checkbox		Show image filter
Edges			
Color	slider	0.0 to 1.0	
Normal	slider	0.0 to 1.0	
Textures			
MIP	slider	-1 to 1.0	

Control	Type	Range or choices	Notes
Freq	slider	-3 to 1.0	
Color	slider	-1 to 1.0	
Bump	slider	-1.0 to 1.0	

Options tab

Control	Type	Range or choices	Notes
Grid Options			
Map	checkbox		Draw Long/Lat lines
Contours	checkbox		Draw Contour lines
Auto	checkbox		Autoset intervals
Grid Options	option buttons	K, C, F	
Lines			
Grid	slider	0.1 to 10	
Miles	color box		
Lat	color box		
Contours	slider	100 to 20000	
Feet	color box		
Text	color box		

Planet

Menu: Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Object tab

Control	Type	Range or choices	Notes
Object			
Name	text		
TidalLock	button		Make the year (orbit) as long as the day and reset the rotation phase, so one side always faces the star.
Temp	display only		
Seasonal	checkbox		
Size(1e3 ml)	slider	1 to 1000	Planet size in thousands of miles.
Grid	slider	1 to 4	
Ht(mls)	slider	0.1 to 20	Maximum terrain height in miles.
Tilt	slider	-90 to 90	Tilt of the rotation axis in degrees; gives the planet seasons.

Control	Type	Range or choices	Notes
Day(hrs)	slider	0 to 200	Length of the planet's day (rotation period) in hours.
Phase	slider	0 to 360	
Orbit			
Radius(x1e6)	slider	1 to 1000	Orbit radius in millions of miles.
Tilt	slider	0 to 90	Inclination of the orbit in degrees.
Year(days)	slider	0 to 1000	Length of the orbit in days.
Phase	slider	0 to 360	
Variability			
Season	slider	0.0 to 1	
Temp	slider	0.0 to 1	

Lighting tab

Control	Type	Range or choices	Notes
Color			
Shine	slider	0.25 to 100	
Albedo	slider	0.01 to 1	
Ambient	color slider		
Specular	color slider		
Diffuse	color slider		
Radiance	color slider		
Shadow	color slider		

Asteroid

Menu: Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Object tab

Control	Type	Range or choices	Notes
Object			
Name	text		
MaxDepth	list	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20	
Smooth	checkbox		Smooth normals
Lines	checkbox		wireframe
Size(miles)	slider	1 to 1000	

Control	Type	Range or choices	Notes
Grid	slider	1 to 4	
Day(hrs)	slider	0 to 200	
Phase	slider	0 to 360	
Shape			
Scale	slider	0.0 to 1	
RNoise	expression field		
Symmetry	slider	0.2 to 1	
VNoise	expression field		
Orbit			
Radius(x1e6)	slider	1 to 1000	
Tilt	slider	0 to 90	
Year(days)	slider	0 to 1000	
Phase	slider	0 to 360	

Lighting tab

Control	Type	Range or choices	Notes
Color			
Shine	slider	0.25 to 100	
Ambient	color slider		
Specular	color slider		
Diffuse	color slider		
Shadow	color slider		

Star

Menu: Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Object tab

Control	Type	Range or choices	Notes
Object			
Name	text		
Type	display only		
Size(mm)	slider	0.1 to 10	
Grid	slider	1 to 4	
Day(hrs)	slider	0 to 200	

Control	Type	Range or choices	Notes
Phase	slider	0 to 360	
Orbit			
Radius(mm)	slider	0 to 100	
Oblique	slider	0 to 0.95	
Year(days)	slider	0 to 1000	
Phase	slider	0 to 360	

Lighting tab

Control	Type	Range or choices	Notes
Color			
Radiance	color slider		
Specular	color slider		
Diffuse	color slider		

Moon

Menu: Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Object tab

Control	Type	Range or choices	Notes
Object			
Name	text		
TidalLock	button		
Temp	display only		
Seasonal	checkbox		
Size(mls)	slider	100 to 10000	
Grid	slider	1 to 4	
Ht(mls)	slider	0.1 to 20	
Tilt	slider	-90 to 90	
Day(hrs)	slider	0 to 200	
Phase	slider	0 to 360	
Orbit			
Radius(x1e3)	slider	1 to 100	
Tilt	slider	0 to 90	
Year(days)	slider	0 to 1000	

Control	Type	Range or choices	Notes
Phase	slider	0 to 360	
Variability			
Season	slider	0.0 to 1	
Temp	slider	0.0 to 1	

Lighting tab

Control	Type	Range or choices	Notes
Color			
Shine	slider	0.25 to 100	
Albedo	slider	0.01 to 1	
Ambient	color slider		
Specular	color slider		
Diffuse	color slider		
Radiance	color slider		
Shadow	color slider		

Corona

Menu: Show, Delete

Properties tab

Control	Type	Range or choices	Notes
Geometry			
Name	text		
Radius	slider	1 to 10	
Color			
Inner	color slider		
Outer	color slider		
Gradient			
Alpha	slider	0.0 to 1	
Internal	checkbox		
Density			
Density	expression field		

Halo

Menu: Show, Delete

Properties tab

Control	Type	Range or choices	Notes
Geometry			
Name	text		
Ht	slider	0.001 to 0.1	
Color			
Inner	color slider		
Outer	color slider		
Gradient			
Alpha	slider	0.0 to 1	

Clouds

Menu: Show, Delete, Save.., Add, Randomize, Default, Replace with.., Remove file..

Object tab

Control	Type	Range or choices	Notes
Orbital			
Name	text		
Orbital	list	2D, 3D, Vol	2D: textured layer, 3D: sprites, Vol: ray-marched volume
DP	checkbox		Double precision noise: smooth noise in close views (2D, Vol), but much slower for Vol
Grid	slider	1 to 8	
Height	slider	10 to 100	
Tilt	slider	0 to 360	
Period	slider	-20 to 20	
Phase	slider	0 to 360	

Lighting tab

Control	Type	Range or choices	Notes
Color			
Shine	slider	0.25 to 100	
Ambient	color slider		
Specular	color slider		
Diffuse	color slider		
Radiance	color slider		

Shape (shown only for some modes) tab

Control	Type	Range or choices	Notes
3D Shape			
3D Shape	expression field		
Top	expression field		
Sprite Properties			
Number	slider	1 to 8	
Scatter	slider	0 to 32	
Size	slider	1 to 128	
Range	slider	1 to 4	
Angle	slider	0 to 1	
Darken	slider	0 to 2	

Images (shown only for some modes) tab

Control	Type	Range or choices	Notes
Sprites			
File	list		
Sprites	list	1x, 4x, 16x	
Type expr	expression field		

Volume (shown only for some modes) tab

Control	Type	Range or choices	Notes
Height			
Ampl	slider	0.5 to 15	
Var	slider	0 to 15	
Shape			
Freq	slider	0.1 to 3	
Erosion	slider	0 to 1	
Density	slider	0.25 to 2	
Darken	slider	0 to 2	
Steps	slider	16 to 256	
Cover	slider	-1 to 1	

Sky

Menu: Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Object tab

Control	Type	Range or choices	Notes
Object			
Name	text		
Grid	slider	1 to 4	
Height	slider	10 to 200	
Opacity	slider	0 to 1	
Pressure	slider	0 to 200	
% GHG	slider	0 to 100	

Lighting tab

Control	Type	Range or choices	Notes
Color			
Halo	color slider		
Sky	color slider		
Twilite	color slider		
Night	color slider		
Scatter	color slider		
Sunset	color slider	0 to 4	
Sharpen			
Sharpen	slider	0.1 to 5	
Sky grad	slider	0.01 to 1	
Twilite grad	slider	0.01 to 1	
Night grad	slider	-0.01 to -1	
Scatter grad	slider	0 to 1	
Redden grad	slider	0.2 to 5	
Haze			
Color	color slider		
Haze	slider	0 to 1	
Min	slider	0 to 5000	
Depth	slider	0 to 100000	

Texture

Menu: Enable, Delete, Save.., Add

Image tab

Control	Type	Range or choices	Notes
Image			
Image	list		
Image	list		
Seasonal	checkbox		
Color			
Ampl	slider	0 to 2.0	
Offset	slider	-2.0 to 2.0	
Color	checkbox		
Bump			
Ampl	slider	-8 to 8	
Offset	slider	0 to 1	
Bump	checkbox		
HtMap			
Ampl	slider	-50 to 50	
Offset	slider	-1 to 1	
HtMap	checkbox		
Overlays			
Start	slider	-1 to 29	
Levels	slider	1 to 25	
Ampl	slider	0.0 to 1.0	
Freq	slider	1.0 to 5.0	

Mapping tab

Control	Type	Range or choices	Notes
Interpolation	option buttons	Nearest, Linear, Mip	
Options			
Clamp	checkbox		
Normalize	checkbox		
Dealias	checkbox		
Triplanar	checkbox		
Modulation			
Modulation	expression field		
Alpha expr	expression field		

Control	Type	Range or choices	Notes
Env. Bias			
Lat	slider	-1 to 1	
Height	slider	-1 to 1	
Bump	slider	-1 to 1	
Slope	slider	-1 to 1	

Sprite

Menu: Enable, Delete, Save.., Add

Distribution tab

Control	Type	Range or choices	Notes
Multilevel			
Levels	list	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	
Size	slider	1 to 200	
Density	expression slider	0.0 to 1.0	
Dispersion			
Size	expression slider	0.0 to 2	
Drop	expression slider	0 to 2	
Density Bias			
Slope	expression slider	-1 to 1	
Hardness	expression slider	-1 to 1	
Latitude	expression slider	-1 to 1	
Height	expression slider	-1 to 1	

Image tab

Control	Type	Range or choices	Notes
Sprites			
File	list		
Sprites	list		
Center	list	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15	
Bias	expression slider	0 to 1	

Plant

Menu: Enable, Delete, Save.., Add, Randomize, Default, Replace with.., Remove file..

Properties tab

Control	Type	Range or choices	Notes
Properties			
Name	text		
Levels	list	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	
Density	expression slider	0.0 to 1.0	
Size			
Max	slider	1 to 200	
Delta	expression slider	0.0 to 2	
Density Bias			
Slope	expression slider	-1 to 1	
Hardness	expression slider	-1 to 1	
Latitude	expression slider	-1 to 1	
Height	expression slider	-1 to 1	
Other			
Drop	expression slider	0 to 2	
Skeleton	checkbox		
Lines	checkbox		
Splines	checkbox		
One	checkbox		

Branch

Menu: Enable, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Properties tab

Control	Type	Range or choices	Notes
Name	text		
Randomize	checkbox		
Levels			
Max	list	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20	
Min	list	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20	
End	checkbox		
First	list	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17,	

Control	Type	Range or choices	Notes
		18, 19, 20	
x10	checkbox		
Later	list	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20	
Size			
Length	expression slider	0.1 to 2	
Taper	expression slider	0.1 to 1	
Width	expression slider	0.1 to 1	
Taper	expression slider	0.1 to 1	
Dispersion			
Divergence	expression slider	0 to 1	
Offset	expression slider	0 to 1	
Random	expression slider	0 to 1	
Density	expression slider	0 to 1	
Curvature			
Amplitude	expression slider	0 to 1	
Direction	expression slider	-1 to 1	

Texture tab

Control	Type	Range or choices	Notes
Image			
File	list		
Image	list		
Enable	checkbox		
Center	expression slider	0 to 1	

Color tab

Control	Type	Range or choices	Notes
Red	expression field		
Green	expression field		
Blue	expression field		
Alpha	expression field		
Color picker	color box		
Revert	button		

Control	Type	Range or choices	Notes
Enable	checkbox		

Leaf

Menu: Enable, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Properties tab

Control	Type	Range or choices	Notes
Name	text		
Cluster	list	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	
X	list	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	
Shape	checkbox		
Randomize	checkbox		
Shape			
Length	expression slider	1 to 50	
Taper	expression slider	0.0 to 1	
Width	expression slider	0.2 to 5	
Taper	expression slider	0.1 to 1	
Dispersion			
Divergence	expression slider	0 to 1	
Offset	expression slider	0 to 1	
Random	expression slider	0 to 1	
Density	expression slider	0 to 1	
Curvature			
Amplitude	expression slider	0 to 1	
Direction	expression slider	-1 to 1	

Texture tab

Control	Type	Range or choices	Notes
Image			
File	list		
Image	list		
Enable	checkbox		
Center	expression slider	0 to 1	

Color tab

Control	Type	Range or choices	Notes
Red	expression field		
Green	expression field		
Blue	expression field		
Alpha	expression field		
Color picker	color box		
Revert	button		
Enable	checkbox		

Color

Menu: Enable, Delete

Color tab

Control	Type	Range or choices	Notes
Red	expression field		
Green	expression field		
Blue	expression field		
Alpha	expression field		
Auto check	checkbox		
Auto	color box		
Revert	button		

Point

Menu: Enable, Delete

Point tab

Control	Type	Range or choices	Notes
X	expression field		
Y	expression field		
Z	expression field		

Gloss

Menu: Enable, Delete

Lighting tab

Control	Type	Range or choices	Notes
Shine	slider	0.5 to 20	
Albedo	slider	0 to 1	

Hardness

Menu: Enable, Delete

Properties tab

Control	Type	Range or choices	Notes
Density	slider	0.0 to 1	

Layer

Menu: Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Layer tab

Control	Type	Range or choices	Notes
Properties			
Label	text		
Square transition slope	checkbox		
Width			
Width	expression slider	0 to 1.0	
Morph			
Morph	expression slider	0 to 1.0	
Drop			
Drop	expression slider	0 to 1.0	
Ramp			
Ramp	expression slider	0 to 1.0	

Map

Menu: Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Properties tab

Control	Type	Range or choices	Notes
Transfer Functions			
Height	expression slider	0 to 1.0	

Control	Type	Range or choices	Notes
Base	expression slider	-1.0 to 1.0	
Scale	expression slider	0.0 to 1.0	
Margin	expression slider	0.0 to 4.0	
Layer Intersection Mode	option buttons	Overlap, Merge	

Fog

Menu: Enable, Delete, Save.., Randomize, Default, Replace with.., Remove file..

Properties tab

Control	Type	Range or choices	Notes
Horizontal Range (feet)			
Min	slider	0 to 1000	
Depth	slider	0 to 10000	
Vertical Range (feet)			
Min	slider	-5000 to 5000	
Depth	slider	0 to 10000	
Properties			
Density	expression field		
Color	color slider		
Glow	slider	0 to 1	

Water

Menu: Enable, Delete, Save.., Randomize, Default, Replace with.., Remove file..

Properties tab

Control	Type	Range or choices	Notes
Object			
Name	text		
Level	slider	-10000 to 10000	
Object	option buttons	Gas, Liquid, Solid	
Auto	checkbox		
Planet temp	display only		
Sky View Reflections	checkbox		Render an extra view above the screen so reflections of off-screen sky and terrain don't fall back to an estimated sky color

Control	Type	Range or choices	Notes
Presets			
Presets	list		
Modulation	expression field		
Default	button		
Volatility			
Evaporation	slider	0.1 to 10	
Sublimation	slider	0.1 to 10.0	

Solid tab

Control	Type	Range or choices	Notes
Surface			
Modulation	expression field		
Default	button		
Mix	slider	0 to 1.0	
Transparency			
feet	color slider	1 to 1000	
Reflection			
%	color slider	0 to 1	
Specular			
shine	slider	0 to 100.0	
albedo	slider	0 to 1.0	
Temperature (C)			
Liquid	slider	-1000 to 1000	
Trans	slider	0 to 10.0	

Liquid tab

Control	Type	Range or choices	Notes
Surface			
Modulation	expression field		
Default	button		
Mix	slider	0 to 1.0	
Transparency			
feet	color slider	1 to 1000	
Reflection			

Control	Type	Range or choices	Notes
%	color slider	0 to 1	
Specular			
shine	slider	0 to 100.0	
albedo	slider	0 to 1.0	
Temperature (C)			
Gas	slider	-1000 to 1000	
Trans	slider	0 to 10.0	

Surface

Menu: Add, Save.., Randomize, Default, Replace with.., Remove file..

Controls tab

Control	Type	Range or choices	Notes
Variables			
Variables	list		
Expr	expression field		
New	button		
Delete	button		
Save	button		

Rocks

Menu: Enable, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Rocks tab

Control	Type	Range or choices	Notes
Properties			
Name	text		
Levels	list	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	
Density	expression slider	0.0 to 1.0	
Size			
Max	slider	0.1 to 0.9999	
Size	list	10, 1, 0.1, 0.01, 0.001, 1e-4, 1e-5, 1e-6, 1e-7	
Delta	expression slider	0.0 to 2.0	
Shape			

Control	Type	Range or choices	Notes
Ampl	expression slider	0.0 to 1.0	
Noise	expression field		
Smooth	checkbox		Smooth normals
Compress	expression slider	0 to 1.0	
Drop	expression slider	0 to 1	
3D	checkbox		
Density Bias			
Slope	expression slider	-1 to 1	
Hardness	expression slider	-1 to 1	
Latitude	expression slider	-1 to 1	
Height	expression slider	-1 to 1	

Rocks 3D

Uses the same tabs as **Rocks**.

Ring

Menu: Show, Delete, Save.., Add

Object tab

Control	Type	Range or choices	Notes
Object			
Name	text		
Grid	slider	1 to 4	
Radius	slider	0.1 to 2	
Width	slider	0.1 to 2	

Lighting tab

Control	Type	Range or choices	Notes
Lighting			
Emission	color slider		
Shadow	color slider		

Snow

Menu: Enable, Delete

Properties tab

Control	Type	Range or choices	Notes
Snow			
Level	slider	-4 to 4.0	
Depth	slider	0.0 to 1.0	
Bias			
Lat	slider	0.0 to 1.0	
Slope	slider	-1 to 1	
Ht	slider	-4 to 4.0	
Bump	slider	-1 to 1.0	

System

Menu: Set View, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Properties tab

Control	Type	Range or choices	Notes
Name	text		
Origin			
X	slider	-10000.0 to 10000.0	
Y	slider	-100.0 to 100.0	
Z	slider	-1000.0 to 1000.0	

View tab

Control	Type	Range or choices	Notes
Set View			
Top	button		
Edge	button		
Oblique	button		

Galaxy

Menu: Set View, Show, Delete, Add, Save.., Randomize, Default, Replace with.., Remove file..

Object tab

Control	Type	Range or choices	Notes
Name	text		
Type	display only		

Control	Type	Range or choices	Notes
Size(ly)	slider	100 to 100000	
Grid	slider	0.25 to 4	
Origin			
X	slider	-100000.0 to 100000.0	
Y	slider	-100000.0 to 100000.0	
Z	slider	-100000.0 to 100000.0	
Rotation			
X	slider	0 to 360	
Y	slider	0 to 360	
Z	slider	0 to 360	

View tab

Control	Type	Range or choices	Notes
Set View			
Core	button		
Rim	button		
Top	button		
Edge	button		
Oblique	button		
Position Radius-Theta-Phi			
R	slider	0 to 100000.0	
T	slider	0 to 360	
P	slider	0 to 360	
Rotation X-Y-Z			
X	slider	0 to 360	
Y	slider	0 to 360	
Z	slider	0 to 360	

Shape tab

Control	Type	Range or choices	Notes
Default Shape			
Spiral	button		
Barred	button		
Globular	button		

Control	Type	Range or choices	Notes
Elyptical	button		
Irregular	button		
Radius			
Core	slider	0 to 1	
Rim	slider	0 to 1	
Twist			
Radius	slider	0 to 2.0	
Angle	slider	0 to 720	
Surface			
In	slider	0 to 1	
Out	slider	0.0001 to 0.1	
Compression			
Vertical	slider	0 to 1.0	
Radial	slider	0 to 1	

Density tab

Control	Type	Range or choices	Notes
Gradient			
Vertical	slider	0 to 2	
Radial	slider	0 to 2	
Distribution			
Diffusion	slider	0.01 to 4	
Scatter	slider	0 to 20	
Noise			
Amplitude	slider	0 to 1	
Vertical	slider	0 to 1	
Saturation	slider	0 to 4	
Cutoff	slider	-1 to 1	
Noise	expression field		
Default	button		

Render tab

Control	Type	Range or choices	Notes
Star Size			

Control	Type	Range or choices	Notes
Near	slider	0 to 5	
Far	slider	0 to 5	
Nova			
Size	slider	0 to 5	
%	slider	0 to 1	
Stars			
Near	slider	10 to 500	
Size	slider	0.5 to 4	
Far	slider	1 to 5	
Dust	slider	0.0 to 4	
Color			
Color	expression field		
Intensity	slider	0 to 1	
Bias	slider	0 to 2	

Fractal

Menu: Enable, Delete

Fractal tab

Control	Type	Range or choices	Notes
Orders			
Start	expression slider	0 to 25	
Levels	expression slider	1 to 25	
Deposition			
Build	expression slider	0.0 to 3.0	
Erode	expression slider	0.0 to 3.0	
Slope Bias			
Magn	expression slider	0 to 3	
Max	expression slider	0.0 to 3.0	
Thresh	expression slider	0 to 1.0	
Mult	expression slider	0.0 to 1.0	
Elevation Bias			
Max	expression slider	0 to 1.0	
Val	expression slider	0 to 1.0	

Control	Type	Range or choices	Notes
Options			
Square	checkbox		
SmoothStep	checkbox		

Erode

Menu: Enable, Delete

Erode tab

Control	Type	Range or choices	Notes
Erosion			
Depth	expression slider	0.0 to 2.0	
Power	expression slider	0.1 to 4.0	
Start	expression slider	-2.0 to 2.0	
Range	expression slider	0.01 to 4.0	
Drainage			
Start	expression slider	0.0 to 20.0	
Orders	expression slider	1.0 to 8.0	
Atten	expression slider	0.0 to 1.0	
Delf	expression slider	1.1 to 4.0	
Amplitude			
Drainage	expression slider	0.0 to 2.0	
Erosion	expression slider	0.0 to 2.0	
Options			
Rounded	checkbox		
Square	checkbox		
SmoothStep	checkbox		

Image Editor

1D tab

Control	Type	Range or choices	Notes
File			
File	list		
File	list	1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024	

Control	Type	Range or choices	Notes
Options			
Norm	checkbox		
Invert	checkbox		
Show Tmps	checkbox		
Mix/Modulate			
Mix/Modulate	color slider		
Mod slider	slider	0.0 to 1.0	
Colors			
Colors	color slider		
Enable	checkbox		
Color Options			
Clamp	checkbox		
Nearest	checkbox		
Reflect	checkbox		
Alpha	checkbox		
>	button		
<	button		

2D tab

Control	Type	Range or choices	Notes
File			
File	list		
File	list	1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024	
File	list	1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024	
File	list	Tile, Decal, Sphere, Cylinder	
Show Tmps	checkbox		
Options			
Norm	checkbox		
Invert	checkbox		
Gray	checkbox		
Color	checkbox		
Options	list		

Control	Type	Range or choices	Notes
Function			
Function	expression field		

Img tab

Control	Type	Range or choices	Notes
File			
File	list		
Tile	checkbox		

Map tab

Same controls as the **Img** tab.

HMap tab

Same controls as the **Img** tab.

Process tab

Control	Type	Range or choices	Notes
File			
File	list		
Grayscale	checkbox		
Tile	checkbox		
Operation			
Operation	list		
Strength	slider	0.0 to 2.0	
Radius	slider	1 to 20	
Iterations	slider	1 to 500	

Function editors

These dialogs open from an expression field: double-click a function token, then right-click and choose **Edit**.

Craters

Craters tab

Control	Type	Range or choices	Notes
Distribution			
Levels	list	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	
Density	expression slider	0.0 to 1.0	
Distribution	list	ID0, ID1, ID2, ID3, ID4, ID5, ID6, ID7, ID8, ID9	
Max-Delta Size			
Max-Delta Size	slider	0.1 to 0.9999	
Max-Delta Size	list	100, 10, 1, 0.1, 0.01, 10-3, 10-4, 10-5, 10-6	
Delta size	expression slider	0.0 to 1.0	
Amplitude			
Depth	expression slider	0.0 to 1.0	
Impact	expression slider	0 to 1.0	
Noise			
Vertical	expression slider	0 to 1.0	
Radial	expression slider	0 to 1.0	
Taper	expression slider	0 to 1.0	
Bias	expression slider	0 to 0.9	

Shape tab

Control	Type	Range or choices	Notes
Geometry			
Rise	expression slider	0.0 to 10.0	
Drop	expression slider	0 to 10.0	
Rim	expression slider	0.0 to 1.0	
Floor	expression slider	0 to 1.0	
Center	expression slider	0.0 to 1.0	
Offset	expression slider	-1 to 1	

Fractal

Fractal tab

Control	Type	Range or choices	Notes
Orders			
Start	expression slider	0 to 25	
Levels	expression slider	1 to 25	
Deposition			
Build	expression slider	0.0 to 3.0	
Erode	expression slider	0.0 to 3.0	
Slope Bias			
Magn	expression slider	0 to 3	
Max	expression slider	0.0 to 3.0	
Thresh	expression slider	0 to 1.0	
Mult	expression slider	0.0 to 1.0	
Elevation Bias			
Max	expression slider	0 to 1.0	
Val	expression slider	0 to 1.0	
Options			
Square	checkbox		
SmoothStep	checkbox		

Noise

Multi-Noise tab

Control	Type	Range or choices	Notes
Orders			
Start	slider	0 to 25	
Levels	slider	1 to 25	
Atten	slider	0 to 1	
DelF	slider	1 to 4	
Homog	slider	-1 to 1	
Shape			
Sqr	checkbox		
Invt	checkbox		
Uns	checkbox		

Control	Type	Range or choices	Notes
Abs	checkbox		
Norm	checkbox		
Scale	checkbox		
LOD	checkbox		
Signal			
Gain	slider	0 to 1	
Offset	slider	-2.0 to 2.0	
Modulation			
Smooth	slider	0 to 1	
Clamp	slider	-0.5 to 2	

Advanced tab

Control	Type	Range or choices	Notes
Noise			
RSeed	list	0, 1, 2, 3, 4, 5	
RMode	list	Vertex, Pixel	
Type	list	Perlin, Voronoi, Simplex	
Animation			
On	checkbox		
Animation	list	1, 0.1, 0.01, 0.001, 1e-4, 1e-5, 1e-6, 1e-7, 1e-8	
Rate	slider	1 to 10	

Erode

Erode tab

Control	Type	Range or choices	Notes
Erosion			
Depth	expression slider	0.0 to 2.0	
Power	expression slider	0.1 to 4.0	
Start	expression slider	0 to 1.0	
Range	expression slider	0.01 to 1.0	
Drainage			
Start	expression slider	0.0 to 20.0	
Orders	expression slider	1.0 to 8.0	

Control	Type	Range or choices	Notes
Atten	expression slider	0.0 to 1.0	
Delf	expression slider	1.1 to 4.0	
Amplitude			
Drainage	expression slider	0.0 to 2.0	
Erosion	expression slider	0.0 to 2.0	
Options			
Rounded	checkbox		
Square	checkbox		
SmoothStep	checkbox		